L&T; SEMICONDUCTOR • ENGINEERING STUDY

REVERSE ENGINEERING OF

MOUSE

A study of the wired optical computer mouse



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From physical components to digital cursor control

REPORT STRUCTURE

This report follows the project outline for product study, product evolution, internal components and working flow. Part II and Part III are organized using the headings shown in the provided reference photograph.

Page	Section	Coverage
3	Part I — Studying the Product	External features, construction, subsystems, modes, purpose and user needs
4	External Features and Dimensions	Physical form, key dimensions and exploded view
5	Product Purpose and User Needs	Use cases, user needs and design considerations
6	Part II — Product Origin to Latest Inventions	Technology evolution across generations
7	Current Status and Engineering Importance	Core technologies and evolution lessons
8	Part III — Internal Components & Working Flow	Product introduction and internal photograph
9	Internal Block Diagram	Functional relationships between subsystems
10	Subsystems — Function and Need	Optical sensing, switches, scrolling, controller and USB
11	Working Principle and Characteristics	How input is converted into USB data
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17	Simulation Circuit and Code	Functional switch-input example
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19	Engineering Discussion	Design trade-offs, maintenance and safety
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Source discipline: Product-level information is taken from the supplied *REVERSE ENGINEERING.docx*. Exact ratings for individual PCB components are not assumed where the source says they cannot be confirmed.

PART I — STUDYING THE PRODUCT

1. External Features

The product studied is identified in the source report as a Dell MS111/MS111-T wired optical mouse. It has a compact plastic body made from an upper shell and a lower base. On the upper side are two primary buttons, a central scroll wheel and a product logo. The lower side accommodates the optical sensing opening, identification label and glide pads. A USB cable is permanently connected to the electronic assembly.

Physical Construct

The shell provides a lightweight protective enclosure while leaving the buttons and wheel accessible. The symmetrical, ambidextrous-style form supports use by either hand. The lower surface is shaped to slide over a desk or mouse pad with low friction.

Subsystems Used

Subsystem	Main role
Mechanical subsystem	Housing, support structure, button travel, wheel support and glide surfaces.
Optical sensing subsystem	Illuminates and observes the surface to estimate mouse movement.
Electronic control subsystem	Processes sensor and switch signals using controller electronics.
User-input subsystem	Detects left-click, right-click and wheel/middle-click actions.
USB power and communication subsystem	Supplies power and carries mouse data to the computer.

Mode of Operation

The mouse operates in wired USB mode. The computer's USB port supplies power. The optical sensor tracks movement; switches detect button presses; and the scroll encoder detects wheel rotation. The controller processes these inputs and reports them to the computer over USB.

EXTERNAL FEATURES AND DIMENSIONS

The provided product image identifies the approximate external dimensions below. These are product-level dimensions, not the dimensions of individual internal components.

Parameter	Approximate value
Length	112.58 mm
Width	63.36 mm
Height	35.5 mm
Weight	68 g
Connection	Wired USB

Exploded View — Parts of a Mouse



The supplied exploded-view illustration separates the top shell, scroll wheel, click-switch assembly, PCB, optical sensor, bottom case and glide pads. It is useful for understanding how the enclosure, user controls and electronics fit together. The exact construction can vary between mouse models.

2. PURPOSE OF THE PRODUCT

The main purpose of the wired optical mouse is to provide a human-computer input interface. It translates hand movement and physical control actions into signals that the computer can use to move the cursor, select items, drag objects, scroll through pages and navigate graphical user interfaces.

3. USER NEEDS ADDRESSED

- **Cursor control:** accurate movement in two dimensions across the screen.
- **Selection:** left-click and right-click actions for choosing and opening items or menus.
- **Scrolling:** wheel rotation for moving through documents and webpages.
- **Middle-click:** scroll-wheel press for applications that support this action.
- **Plug-and-play use:** USB connection allows the mouse to operate without a separate battery.
- **Reliable wired communication:** data travels over the cable without a wireless pairing step.
- **Low maintenance:** optical tracking removes the mechanical ball and rollers used in older mice.
- **Simple operation:** familiar buttons and wheel require little training.

Design Considerations

A useful mouse balances tracking performance, comfortable shape, button feel, cable durability, manufacturing cost and ease of assembly. The low-profile enclosure helps the user hold and move the device, while the glide pads reduce friction. The cable avoids battery charging but limits the range of movement.

Product boundaries

The source describes a basic wired optical model with approximately 1000 DPI tracking, two primary buttons and a scroll wheel with middle-click. It does not include Bluetooth or a wireless module. These limits are important when comparing it with more advanced modern products.

PART II — PRODUCT ORIGIN TO LATEST INVENTIONS

1. Difference in Technology Across Versions

The mouse has evolved from mechanical movement detection to optical tracking and, later, to high-resolution wired and wireless products. Each stage improved some aspects of usability while introducing different costs and limitations.

1.1 Early Stage — Mechanical Ball Mouse

Early computer mice used a ball on the underside. As the mouse moved, the ball rotated rollers inside the enclosure. These rollers converted movement along two axes into signals that could be interpreted by the computer.

Improvements: It enabled basic two-dimensional cursor control and could be manufactured at relatively low cost.

Drawbacks: The ball and rollers collected dust and dirt, which could reduce tracking accuracy. Moving parts wore over time and often needed cleaning.

1.2 First Generation — Optical Mouse

Optical mice replaced the mechanical tracking ball with an LED and optical sensor. The sensor observes surface texture changes as the device moves, reducing dependence on rollers and other tracking mechanisms.

Improvements: No tracking ball, less mechanical wear, lower maintenance and improved reliability.

Drawbacks: Early optical designs could struggle on transparent, reflective or otherwise unsuitable surfaces.

PART II — TECHNOLOGY EVOLUTION (CONTINUED)

1.3 Second Generation — Wired USB Optical Mouse

The Dell MS111/MS111-T described in the source is a simple USB wired optical mouse. It combines optical motion sensing, electronic signal processing and USB communication in a compact enclosure.

- Optical tracking instead of a mechanical ball.
- USB plug-and-play connectivity and power.
- No battery replacement or charging.
- Simple construction and reliable wired communication.
- Low maintenance for normal use.

Drawbacks: The cable restricts movement, a USB connection is required, and the fixed resolution of approximately 1000 DPI offers fewer adjustment options than many modern models.

1.4 Latest Generation — Modern Smart/Wireless Mouse

Modern products may include high or adjustable DPI, Bluetooth or 2.4 GHz wireless communication, rechargeable batteries, programmable buttons, multi-device switching and software-based customization. USB-C charging is also common in some rechargeable models.

Improvements: More portability, customization, tracking flexibility and connectivity choices.

Drawbacks: These features can increase cost and electronic/software complexity. Wireless models also need battery charging or replacement.

2. Current Status — Core Smart Technologies Used in Product

- Optical motion sensing.
- Integrated electronic signal processing.
- USB Human Interface Device (HID) communication.
- Microswitch-based button detection.
- Rotary encoder-based scrolling.
- USB power supply and compact PCB construction.

PART II — ENGINEERING IMPORTANCE OF THIS EVOLUTION

3. Engineering Importance of This Evolution

The evolution from mechanical-ball mice to optical and modern wireless mice demonstrates progress in accuracy, reliability, usability and miniaturization. Replacing mechanical tracking parts with optical sensing reduced wear and maintenance. USB simplified power and data communication, while wireless technologies increased portability and added configuration options.

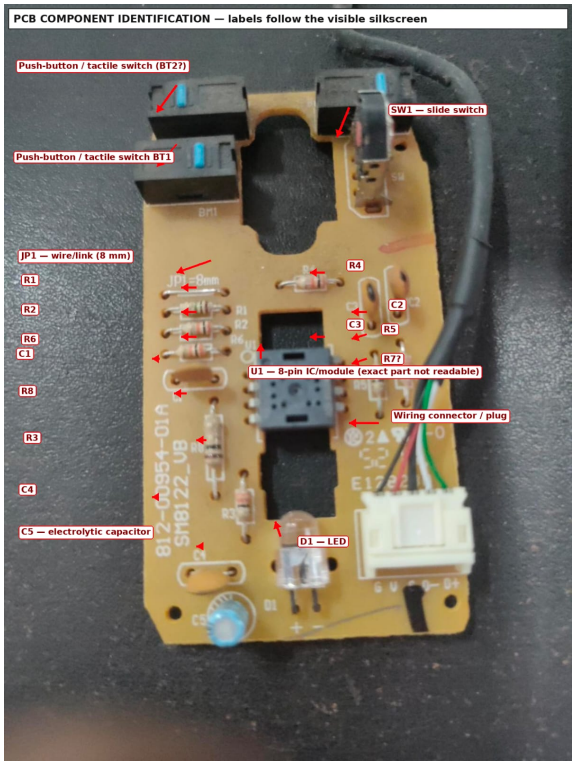
The mouse is an example of multidisciplinary product engineering. Its performance depends on the mechanical shell, optical sensor, electronic controller, switch mechanics, cable and communication protocol working together. A change to one subsystem can affect cost, ergonomics, reliability or compatibility.

PART III — INTERNAL COMPONENTS & WORKING FLOW

1. Introduction of the Product

The wired optical mouse contains a printed circuit board (PCB), optical sensing components, controller electronics, microswitches, a scroll-wheel mechanism, resistors, capacitors and a USB cable connection. The PCB provides the electrical platform that connects the components.

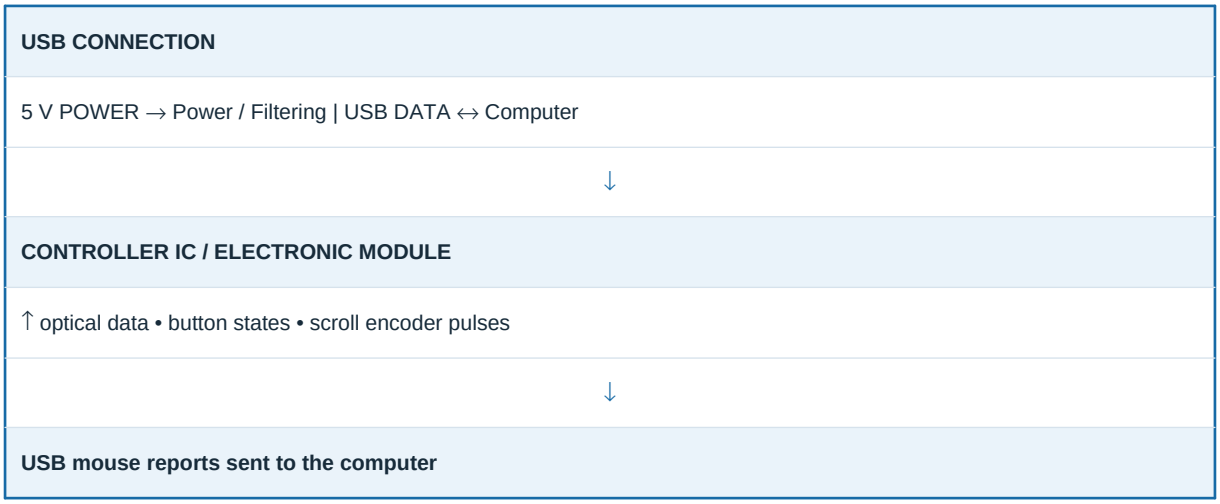
2. Photograph of the Internal Components of the Product



The provided PCB photograph includes annotations for the visible switches, slide switch, resistors, capacitors, an 8-pin IC/module, LED and wiring connector. The labels follow the visible silkscreen where shown; the exact part number of the IC is not readable in the supplied image.

PART III — INTERNAL BLOCK DIAGRAM (FUNCTIONAL VIEW)

The diagram below summarizes the functional relationship between the USB connection, controller, user inputs and optical sensing subsystem. It is a conceptual block diagram rather than a verified circuit schematic.



Inputs to the controller

Input block	Information provided
Optical sensor and LED	Surface-image changes used to estimate movement direction and amount.
Left/right switches	Digital state changes when either primary button is pressed or released.
Scroll encoder	Pulses or transitions corresponding to wheel rotation.
Middle-click switch	A switch state generated when the wheel is pressed down.

The exact electrical arrangement depends on the specific PCB revision. The source material does not provide a verified schematic or confirmed controller part number, so this diagram should be read as a functional view only.

PART III — INTERNAL SUBSYSTEMS WITH DETAILED EXPLANATION

4.1 Function of the Subsystem — Why Is It Used?

A. Optical Sensing Subsystem

The optical LED illuminates the surface beneath the mouse. The optical sensor captures changes in the surface pattern as the mouse moves. The controller uses these changes to estimate movement.

B. Switch/Input Subsystem

The left and right microswitches detect primary button presses. A switch under the scroll wheel detects middle-click. These switches turn a physical press into an electrical input that the controller can read.

C. Scroll Subsystem

The scroll wheel is connected to a rotary encoder. As the wheel rotates, the encoder generates electrical transitions that allow the controller to identify wheel movement and direction.

D. Controller Subsystem

The controller IC receives movement and switch information, interprets the signals and prepares mouse data for the USB interface. The exact IC identity is not confirmed by the supplied PCB photograph.

E. Power and Communication Subsystem

USB supplies approximately 5 V DC and provides a data path to the computer. Power filtering and the PCB's conductive tracks support stable operation of the electronic parts.

F. Mechanical Subsystem

The plastic shell, wheel support, button structures and glide pads keep the parts aligned and allow the mouse to move smoothly across a surface.

PART III — WORKING PRINCIPLE AND CHARACTERISTICS

4.2 Working Principle

When the mouse is connected to a computer, USB power energizes the PCB and optical system. The LED illuminates the surface and the optical sensor observes changes in the surface pattern. As the mouse moves, the controller calculates the direction and amount of movement from the sensor information.

When the user presses a button, the corresponding microswitch changes electrical state. Rotation of the scroll wheel generates signals through the rotary encoder. The controller combines movement, button and scroll information and sends the resulting data to the computer through USB. The operating system converts that information into cursor movement, clicks and scrolling.

4.3 Characteristics

Characteristic	Description
Connection	Wired USB
Tracking method	Optical
Resolution	Approximately 1000 DPI, as stated in the source report
Buttons	Two main buttons plus scroll-wheel/middle-click functionality
Power	USB-powered; approximately 5 V DC
Battery	Not required
Wireless / Bluetooth	Not included in the described model
Construction	Compact PCB and lightweight plastic housing
Maintenance	Low under normal use; no tracking ball to clean

4.4 Role or Need to Use It in This Product

Each subsystem is essential to convert physical user actions into digital commands. The optical system detects motion, switches capture button actions, the encoder detects scrolling, the controller processes signals, and USB provides power and communication.

PART III — COMPLETE WORKING FLOW (STEP-BY-STEP)

5. Complete Working Flow

Step	Operation
1	The mouse is connected to the computer through USB.
2	The USB port supplies approximately 5 V power to the mouse.
3	The PCB, controller and optical system become powered.
4	The optical LED illuminates the surface below the mouse.
5	The optical sensor continuously observes the surface pattern.
6	Mouse movement changes the pattern seen by the sensor.
7	The controller calculates movement information.
8	A button press changes the state of the corresponding microswitch.
9	Scroll-wheel rotation causes the rotary encoder to generate signals.
10	The controller processes movement, button and scroll inputs.
11	The processed information is transmitted through USB.
12	The computer interprets the data and moves the cursor, clicks or scrolls.

Functional summary

> *Physical action → sensor or switch signal → controller processing → USB data report → computer response.*

This flow is intended to explain the product's functional operation. It does not replace a circuit-level schematic, firmware listing or USB protocol trace for the exact device.

PART III — COMPONENTS LIST / BILL OF MATERIAL

6. Components List — Bill of Material

S. No.	Component	Quantity	Function
1	PCB	1	Supports and electrically connects the components.
2	Controller IC/module	1	Processes sensor and button signals and manages communication.
3	Optical sensor	1	Detects mouse movement.
4	Optical LED	1	Illuminates the surface for optical sensing.
5	Left-click microswitch	1	Detects left-click input.
6	Right-click microswitch	1	Detects right-click input.
7	Scroll-wheel encoder	1	Detects wheel rotation.
8	Scroll-wheel click switch	1	Detects middle-click input.
9	Resistors	Several	Current limiting and signal conditioning.
10	Capacitors	Several	Filtering and voltage stabilization.
11	USB cable	1	Provides power and data communication.
12	Plastic housing	1 set	Mechanical protection and support.

Visual identification note

The annotated PCB image also marks components such as BT1/BT2 tactile switches, SW1 slide switch, JP1 wire/link, an 8-pin IC/module, LED D1 and several R/C reference designators. Because the image does not establish every component's electrical value or exact part number, these should not be inferred from appearance alone.

PART III — COMPONENT RATINGS & SPECIFICATIONS

7. Component Ratings & Specifications

Parameter	Specification stated in source
Product	Dell MS111/MS111-T
Part number	X9DCG
Product type	Wired optical mouse
Connection	USB
Tracking	Optical
Resolution	Approximately 1000 DPI
Main buttons	2
Scroll wheel	1
Middle click	Yes
Power supply	USB, approximately 5 V DC
Battery	Not required
Wireless	No
Bluetooth	No
Approximate dimensions	112.58 × 63.36 × 35.5 mm
Approximate weight	68 g
PCB	Compact electronic control board

Important qualification

The source report explicitly notes that exact electrical ratings of individual resistors, capacitors and the controller IC cannot be confirmed from the visible markings in the photograph. No resistor values, capacitor values, IC model number, current draw or detailed sensor specifications are therefore assigned here.

The dimensions and weight are approximate product-level figures supplied in the source. They should not be treated as manufacturing tolerances.

PART III — COMPONENT FUNCTION TABLE

8. Component Function Table

Component name	Function	Specifications / identification
Controller IC/module	Processes mouse input and controls USB communication.	Integrated controller; exact part number not readable.
Optical sensor	Detects movement relative to the surface.	Optical tracking; exact sensor model not supplied.
Optical LED (D1 in photo)	Illuminates the surface for optical sensing.	LED visible in the annotated PCB image.
Left microswitch	Detects left-button press.	Mechanical switch.
Right microswitch	Detects right-button press.	Mechanical switch.
Scroll encoder	Detects wheel rotation and direction.	Rotary encoder.
Middle-click switch	Detects pressing the wheel downward.	Mechanical switch.
Capacitors	Filter noise and stabilize supply voltage.	Values vary by circuit; not confirmed.
Resistors	Control current and signal levels.	Values vary by circuit; not confirmed.
USB cable	Provides power and transfers data.	Wired USB connection.
PCB	Provides electrical connections and component support.	Printed circuit board.
Plastic housing	Protects and supports internal parts.	Upper shell and bottom case.
Glide pads	Reduce friction during movement.	Low-friction pads on the base.

The component table describes functional roles from the source report and the supplied visual labels. Where the image does not clearly reveal a value or part identity, the table deliberately records that limitation.

PART IV — SIMULATION & RESULTS

1. Circuit Diagram of the Product

A complete commercial optical mouse includes a dedicated optical sensor and controller IC. Reproducing the full circuit exactly requires the component part numbers and the original PCB schematic. Those details are not available in the supplied source material.

For a basic simulation, the button-input subsystem can be represented with a push switch, a pull-up or pull-down arrangement and a controller input. This demonstrates the principle of detecting a button press, but it does not simulate the complete optical tracking or USB communication system.

Functional element	Role in simplified circuit
Supply	Represents the logic supply available to the input circuit.
Push switch	Changes the input state when pressed.
Input pin	Reads the switch state as a digital HIGH or LOW.
Resistor / internal pull-up	Prevents the input from floating when the switch is open.
LED indicator (optional)	Makes the input state visible during demonstration.

Scope and limitation

The simulation is a teaching model of switch detection. It should not be presented as the exact schematic of the Dell MS111/MS111-T, and it does not validate the optical sensor, controller firmware or USB HID implementation.

PART IV — SIMULATION RESULTS AND OBSERVATIONS

Simulation Result

When the push button is pressed, the controller or microcontroller detects the change in input and the LED turns ON. When the button is released, the input returns to its inactive state and the LED turns OFF. This demonstrates the basic principle used by mechanical mouse switches to generate digital input signals.

Test condition	Expected input	Expected output
Button released	HIGH with INPUT_PULLUP	LED OFF
Button pressed	LOW	LED ON
Button released again	HIGH	LED OFF

Observations

- A digital input needs a defined state; the internal pull-up avoids a floating input.
- The input logic is active-low because pressing the switch connects the input to ground in the typical INPUT_PULLUP arrangement.
- A simple switch simulation verifies the input concept only; it does not verify the full mouse PCB.
- The actual optical mouse controller integrates sensor processing and USB communication beyond this demonstration.

Evidence to add before submission

- A screenshot showing the actual simulated circuit.
- A link to the group's recorded demonstration, if required by the assignment.
- A short note naming the simulation tool and any component settings used.

ENGINEERING DISCUSSION

Design Trade-offs

Design choice	Benefit	Trade-off
Wired USB connection	Power and data through one cable; no battery.	Cable limits range and can snag.
Optical tracking	No ball or tracking rollers; low maintenance.	Performance depends on surface and sensor design.
Compact PCB	Small enclosure and integrated functions.	Repair and component identification can be difficult.
Simple button layout	Familiar, easy-to-use input controls.	Fewer programmable features than advanced mice.
Plastic housing	Lightweight and economical.	Durability depends on material and assembly quality.

Reliability and Maintenance

Routine care includes keeping the optical opening and glide surfaces clean, avoiding sharp pulls on the USB cable, and preventing liquid from entering the enclosure. If the mouse is opened, the small switches, wheel mechanism and cable connector should be handled carefully to avoid damaging solder joints or plastic clips.

Safety and handling

The device is powered from a low-voltage USB supply during normal use. Do not connect an opened or damaged PCB to a computer if exposed conductors may short. The report is an educational reverse-engineering study and does not recommend modifying the mouse while it is powered.

Limitations of the study

The source material does not include a verified complete schematic, exact controller part number, measured component values, USB traces or laboratory measurements. As a result, the functional explanation is grounded in the provided report and visual evidence, while undocumented electrical details remain unspecified.

CONCLUSION

Final Conclusion

The reverse engineering of the Dell MS111/MS111-T wired optical mouse demonstrates how a familiar computer peripheral combines mechanical design, optical sensing, electronic control and digital communication. The outer shell supports comfortable use, while the internal PCB connects the switches, optical system, controller and USB cable.

The shift from mechanical-ball tracking to optical sensing reduced maintenance and improved reliability. USB provides both power and a communication path, while the controller converts movement, button presses and scrolling into data the computer can interpret. Studying the mouse therefore gives practical insight into sensors, switches, PCBs, subsystem integration and human-computer interaction.

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Note: Confirm the group-specific details and add the actual simulation screenshot/video link before final academic submission.